

Comprehensive Plan for KINSING Multi-Carrier Tracking System

1. Objectives and High-Level Vision

KINSING's tracking portal should provide customers with a **simple public page** where they can enter a KINSING tracking number and see the status of their shipment. Customers will not need to create an account, but they will see key shipment details (sender, receiver, origin/destination, weight/pieces and current status) and a timeline of status updates. The system must integrate information from multiple carriers (initially **DPD** for ground shipping in Germany and **Lufthansa Cargo** for air freight) and translate each carrier's raw events into a **unified set of KINSING statuses**. Only non-sensitive information will be shown on the public page; all carrier details (DPD tracking numbers, Lufthansa AWB numbers, pricing etc.) will be hidden and stored in the backend.

Why multi-carrier integration? The **Anchanto multicarrier guide** explains that a tracking system should integrate with several shipping carriers and **standardize the raw tracking data** received from them; this ensures end-to-end tracking across different carriers ¹. It also notes that parcel tracking systems often implement **data standardization** to ensure data consistency and reliability ². Implementing these practices will allow KINSING to consolidate updates from DPD and Lufthansa into a single clear status timeline for the customer.

2. System Architecture

1. Public Tracking Interface (Web):

2. A simple responsive web page (`kinsing.de/track`) where customers enter the KINSING tracking number.
3. Displays unified status timeline, sender/receiver information, origin/destination, weight and pieces, last update time and next expected milestone.
4. Does **not** display DPD or Lufthansa tracking numbers, internal notes, pricing or personal data beyond the sender/receiver names and city/area.

5. Backend Service:

6. API integration layer that connects to carrier APIs.

- **DPD API:** Cargoson's DPD API offers shipment booking, **tracking and status updates** ³; it requires an API key and uses bearer token authentication ⁴. KINSING will use the tracking endpoint to pull current status and map it to internal statuses.
- **Lufthansa Cargo API:** The Lufthansa Developer Portal lists a **"Shipment Tracking"** resource that retrieves tracking updates for LH Cargo shipments ⁵. KINSING will query this API for each AWB associated with a shipment and map the returned milestones.

7. **Status-mapping engine** that translates carrier events (e.g., RCS, DEP, ARR, NFD) into KINSING statuses (e.g., "Received at warehouse", "Departed Germany", "Arrived Gambia", "Ready for pickup"). Air cargo codes are defined in logistics glossaries:
 - **RCS** means cargo and documents are received "Ready for Carriage" ⁶ ;
 - **DEP** indicates cargo and documents **departed at the airport of origin** ⁷ ;
 - **ARR** means cargo and documents **arrived at the airport of destination** ⁸ ;
 - **NFD** means cargo and documents are **ready for pick-up** and the customer (forwarder) has been notified ⁹ .
8. **Database layer:** Use PostgreSQL to store shipments, customers, carriers, status history and user audit logs. The database will assign a unique KINSING tracking number to each shipment and relate it to multiple carrier references.
9. **Notification service:** In later phases, integrate with messaging providers (WhatsApp, email, SMS) to send automatic status updates to customers.
10. **Admin Dashboard:**
11. Secured login for KINSING staff.
12. Functions to create shipments, assign DPD shipping labels and Lufthansa AWB numbers, update or override statuses, search shipments and view logs.
13. Role-based access control to ensure only authorized employees can perform certain actions (e.g., editing shipments versus viewing only).
14. Bulk import/export features and printing of labels or manifests.
15. **Security Considerations:**

A robust tracking system must protect data and remain resilient against attacks. Splunk's website security guide highlights foundational measures:

- **Install SSL certificates** to encrypt data between the site and its visitors ¹⁰ . All communications (public and admin) must use HTTPS.
 - **Use strong passwords and two-factor authentication** for the admin portal ¹¹ . Avoid default or shared accounts.
 - **Limit user permissions** so that only necessary staff have administrative access ¹² .
 - **Implement anti-bot protections** such as CAPTCHA to prevent automated credential stuffing attacks ¹³ .
 - **Regularly backup your data** and choose a secure hosting provider with firewalls and DDoS protection ¹⁴ .
- Advanced techniques like a web application firewall (WAF), content security policies, intrusion detection systems, DNSSEC and IP whitelisting further strengthen defenses ¹⁵ . These measures should be incorporated during deployment and operations.

3. Detailed Requirements & Features

3.1 Public Tracking Page (MVP)

1. **Tracking search:** An input field where customers enter a KINSING tracking number. The system queries the backend and returns shipment details. Use simple but hard-to-guess tracking numbers (e.g., `KIN-26-A7X9P2`) to prevent enumeration.

2. Shipment summary: Show:

3. KINSING tracking number.

4. Sender and receiver names.

5. Origin city/country and destination city/country.

6. Weight and number of pieces.

7. Current status and last update date/time (in Europe/Berlin timezone).

8. **Optional:** Pickup or delivery address (street and town) if KINSING chooses to display this; avoid disclosing phone numbers or full contact details.

9. **Status timeline:** A chronological list of events. Each event displays the date/time and a user-friendly description. For example:

Carrier code	Meaning (C2K milestone)	KINSING status
RCS	Cargo and documents are received "Ready for Carriage" and accepted by the airline ⁶	Received at KINSING warehouse
DEP	Cargo and documents departed at airport of origin ⁷	Departed Germany
ARR	Cargo and documents arrived at destination airport ⁸	Arrived in Gambia
NFD	Cargo and documents ready for pick-up; consignee notified ⁹	Ready for pickup

DPD statuses (e.g., "In transit," "Delivered") will be similarly mapped to KINSING stages such as "On the way to KINSING," "Received at KINSING," "Prepared for air freight" and "Delivered."

1. **Search results caching:** To reduce API calls and improve performance, store the latest carrier status in the database and only refresh when the data is older than a defined threshold (e.g., 15 minutes or when the customer refreshes the page).
2. **Error handling:** Display user-friendly messages for invalid or unknown tracking numbers. Provide contact information (e.g., KINSING customer service email or phone) for assistance.
3. **Localization & timezone:** Display dates in the user's locale (German/English) and convert timestamps to Europe/Berlin timezone.

3.2 Admin Dashboard

1. **Authentication & Authorization:** Only authorized employees can access the dashboard. Use a secure authentication system with role-based permissions (e.g., admin vs. read-only). Implement strong passwords and 2FA ¹¹.
2. **Shipment management:** Create new shipments with sender/receiver details, route (Germany → Gambia or Gambia → Germany), weight, pieces, and optional notes. The system generates a unique KINSING tracking number. Staff can attach carrier references (DPD tracking number, Lufthansa AWB) to the shipment.
3. **Status updates:**
4. **Automatic:** The backend queries carrier APIs regularly (e.g., hourly or via webhook/push where supported) and updates internal statuses.

5. **Manual override:** Staff can override or add notes if there are discrepancies or exceptional situations.
6. **Search & filters:** Search by KINSING tracking number, customer name, phone, status, route, date range etc. Use filters to quickly identify shipments needing attention.
7. **Notifications:** Option to send WhatsApp, SMS or email updates to customers. Provide a template editor for status messages.
8. **Reporting:** Export shipments, generate manifests, view statistics (e.g., number of packages in transit, delivered, average transit time). Provide dashboards for management insights.
9. **Carrier management:** Manage API credentials (DPD API keys, Lufthansa tokens). Set polling intervals and view API call logs.

3.3 Backend & Integration

1. **Carrier API wrappers:** Build modular services for each carrier:
2. **DPD Module:** Use the Cargoson API for tracking; authentication requires a bearer token ⁴. A scheduled job fetches latest status for each DPD shipment. Keep track of rate limits and error handling.
3. **Lufthansa Module:** Use the Lufthansa Cargo "Shipment Tracking" API ⁵; authentication is via OAuth2 (client credentials). Implement polling or subscription (if supported) for updates.
4. For future carriers (e.g., Brussels Airlines, UPS, DHL), implement additional modules without changing the core logic.
5. **Status normalization:** Each carrier returns events using its own codes. Normalize these events into internal statuses (e.g., "registered," "in transit," "air freight booked," "arrived," "ready for pickup," "delivered"). Use the mapping table above and extend as needed.
6. **Database schema:**
7. `shipments` table: id, kinsing_tracking, sender_name, receiver_name, origin_city, destination_city, weight, pieces, route, created_at, current_status, last_update.
8. `carrier_references`: id, shipment_id (FK), carrier_name, carrier_tracking_number (DPD number or AWB), last_carrier_status, last_carrier_timestamp.
9. `status_history`: id, shipment_id (FK), status_code (internal), description, event_time, carrier_event_code (optional).
10. `users`: id, username, password_hash, role.
11. Additional tables for notifications, logs, carriers etc.
12. **API server:** Build the backend using Node.js (e.g., NestJS) or another preferred framework. Provide REST endpoints for the public page (GET /track?code=), admin operations (CRUD shipments, search, update), and webhooks (for carriers that support push). Implement server-side caching and rate limiting.
13. **Infrastructure & hosting:** Deploy the frontend on a platform like Vercel and the backend on a cloud service (e.g., AWS, DigitalOcean). Use environment variables for API keys and secrets. Connect to a managed PostgreSQL database (e.g., Supabase or AWS RDS).
14. **Logging & monitoring:** Log all API requests, responses, and errors. Set up monitoring and alerts for carrier integration failures. Regularly audit logs to detect suspicious activities.

4. Development Phases & Milestones

1. **Planning & Requirements (Week 1)**

2. Finalize system requirements (features, data fields, user roles, compliance with GDPR).

3. Register for DPD and Lufthansa Cargo API credentials.

4. Design & Architecture (Week 2)

5. Define database schema and status mapping.

6. Create API specifications for internal endpoints.

7. Design UI mockups for the public page and admin dashboard.

8. Backend Development (Weeks 3-5)

9. Implement the database and data models.

10. Build carrier integration modules for DPD and Lufthansa.

11. Develop the status normalization engine.

12. Create REST endpoints for tracking queries and admin operations.

13. Frontend Development (Weeks 5-6)

14. Build the public tracking page using Next.js or React.

15. Build the admin dashboard with secure login, forms for shipment creation and tables for management.

16. Ensure responsive design and accessibility.

17. Security Implementation (Weeks 6-7)

18. Set up HTTPS with SSL certificates ¹⁰.

19. Implement authentication, authorization and 2FA ¹¹.

20. Configure WAF, CSP, intrusion detection, backups and anti-bot protections ¹⁶ ¹⁵.

21. Testing & Quality Assurance (Week 7)

22. Unit tests for API endpoints, carrier integrations and status mapping.

23. Integration tests for end-to-end flows (search, updates, error handling).

24. Security penetration testing and vulnerability scanning ¹⁷.

25. User acceptance testing with sample data and real shipments.

26. Deployment & Launch (Week 8)

27. Deploy backend and database; set environment variables for API keys.

28. Deploy the public tracking site and admin dashboard.
29. Migrate existing shipments into the system and generate new KINSING tracking numbers.
30. Train staff on using the dashboard.

31. **Post-Launch Enhancements**

32. Add WhatsApp/SMS/email notifications for status changes.
33. Implement push notifications using Lufthansa's "Shipment Tracking Subscribe" API ⁵ (if available).
34. Integrate additional carriers (e.g., Brussels Airlines, DHL).
35. Develop customer accounts (optional) to view multiple shipments.
36. Offer mobile-friendly or PWA (Progressive Web App) features.

5. Compliance and Data Protection

Operating between Germany and Gambia requires adherence to data protection regulations such as the **EU's General Data Protection Regulation (GDPR)**. The system must:

1. Collect and store only necessary personal data (sender/receiver names, addresses, phone numbers). Do not expose phone numbers or sensitive details on the public page.
2. Provide a privacy notice explaining what data is collected and how it is used.
3. Allow customers to request removal or correction of their personal data.
4. Encrypt personal data at rest and in transit (via SSL).
5. Restrict access to personal data to authorized staff members and enforce role-based permissions ¹².

6. Review and Quality Assurance

This plan has been **reviewed multiple times** to ensure completeness, consistency and simplicity:

1. **Initial Draft:** Outlined high-level objectives, system architecture and features based on user requirements. Verified that multi-carrier integration and status normalization are essential and supported by credible sources ¹.
2. **Secondary Review:** Expanded sections on backend integration, database design and security. Added citations for carrier API capabilities ³ ⁵ and air cargo status definitions ⁷ ⁹. Included a mapping table for status codes and internal statuses. Emphasized data standardization as recommended by logistics industry guides ².
3. **Final Review:** Refined the development phases, added compliance considerations (GDPR), and supplemented security best practices from Splunk's article (SSL, strong authentication, anti-bot measures, backups, WAF, IDS) ¹⁸ ¹⁵. Ensured the plan is structured logically with clear headings and bullet points and remains simple enough for stakeholders to understand.

7. Conclusion

By following this comprehensive plan, KINSING can build a **professional and user-friendly multi-carrier tracking system** that integrates DPD and Lufthansa Cargo data, delivers clear status updates to customers without exposing sensitive information, and provides a secure and efficient

admin interface for staff. The plan emphasises data standardization, security best practices and phased implementation, ensuring that the solution is scalable and compliant with regulations.

1 2 **What is a Multicarrier Parcel Tracking System?**

<https://anchanto.com/what-is-a-multicarrier-parcel-tracking-system/>

3 4 **DPD PL API Documentation - Shipping, Tracking & Freight API Integration | Cargoson**

<https://www.cargoson.com/integrations/dpd-polska-sp-z-o-o/api-documentation>

5 **Lufthansa Developer Center - Cargo**

https://developer.lufthansa.com/docs/read/api_details/cargo/Cargo

6 7 8 9 **Air Cargo Terms and Abbreviations - United Worldwide Logistics**

<https://unitedwl.com/services/air-cargo-terms-abbreviations/>

10 11 12 13 14 15 16 17 18 **Website Security: Tips & Best Practices for Securing Websites | Splunk**

https://www.splunk.com/en_us/blog/learn/website-security.html